

Capstone Project: Smart Training Support Intelligence System

Objective

Build a complete NLP-based support assistant for an online training company that can:

- Preprocess support messages
- Analyze important words using TF-IDF
- Train simple Word2Vec embeddings
- Find similar support tickets
- Classify ticket category
- Predict sentiment
- Extract named entities
- Use an RNN classifier
- Answer FAQs using Simple RAG without LLM
- Prepare prompt templates

Dataset 1: Support Tickets

ticket_id	message	category	sentiment
1	I cannot access the course	Course Access	Negative
2	The trainer explained the concepts very well	Feedback	Positive
3	When is the next live class	Schedule	Neutral
4	My certificate name is incorrect	Certificate	Negative
5	The workshop was excellent	Feedback	Positive
6	I am unable to login to the LMS	Login Issue	Negative
7	Please share the recording link	Recording	Neutral
8	The course content is very useful	Feedback	Positive
9	I forgot my LMS password	Login Issue	Negative
10	How can I download my certificate	Certificate	Neutral
11	The live class was very helpful	Feedback	Positive
12	I missed yesterday's session	Schedule	Neutral

Dataset 2: Knowledge Base

```
kb_id,question,answer
```

```
1,How to access the course,Login to the LMS using your registered email a
2,How to get certificate,Certificates are issued after completing the rec
3,Where are recordings available,Recordings are available inside the cour
4,When are live classes conducted,Live classes are conducted as per the a
5,How to reset password,Use the forgot password option on the LMS login p
```

Module 1: Text Preprocessing

Students must:

```
Convert text to lowercase
Remove punctuation
Tokenize text
Count words
```

Output:

```
Original Message:
I cannot access the course
```

```
Tokens:
['i', 'cannot', 'access', 'the', 'course']
```

```
Word Count:
5
```

Module 2: TF-IDF Analysis

Use support ticket messages.

Tasks:

```
Create TF-IDF matrix
Print vocabulary
Display TF-IDF scores
Identify important words per ticket
```

Example:

```
Ticket:
I cannot access the course
```

```
Important Words:
access
course
cannot
```

Module 3: Word2Vec Embeddings

Train Word2Vec using ticket messages.

Tasks:

```
Train Word2Vec
Print vector for "course"
Print vector for "certificate"
Find similar words
Compare word similarities
```

Examples:

```
course vs lms
certificate vs download
recording vs link
trainer vs class
```

Module 4: Cosine Similarity Ticket Search

Build a similar-ticket finder.

Input:

```
I cannot login
```

Expected output:

```
Top Similar Tickets:
1. I am unable to login to the LMS
2. I forgot my LMS password
3. I cannot access the course
```

Concepts used:

```
TF-IDF
Cosine Similarity
Top-k retrieval
```

Module 5: Text Classification

Build a ticket category classifier.

Categories:

```
Course Access
Login Issue
Certificate
Recording
Schedule
Feedback
```

Model:

```
Token IDs
↓
Embedding Layer
↓
Average Embeddings
```

↓
Linear Layer
↓
Category Prediction

Example output:

Message:
My certificate name is incorrect

Predicted Category:
Certificate

Module 6: Sentiment Analysis

Build a sentiment classifier.

Classes:

Positive
Negative
Neutral

Example output:

Message:
The workshop was excellent

Predicted Sentiment:
Positive

Model can use:

Embedding Layer
Average Embedding
Linear Layer
CrossEntropyLoss
Adam Optimizer

Module 7: Named Entity Recognition

Use spaCy NER.

Sample inputs:

```
Rahul attended the workshop in Hyderabad.  
Anita completed the Python course in June 2025.  
The session was conducted in Bengaluru.
```

Expected output:

```
Rahul → PERSON  
Hyderabad → GPE  
Anita → PERSON  
June 2025 → DATE  
Bengaluru → GPE
```

Module 8: RNN-Based Classifier

Build an RNN model for ticket category classification.

Architecture:

```
Token IDs  
↓  
Embedding Layer  
↓  
RNN  
↓  
Final Hidden State  
↓  
Linear Layer  
↓  
Category Prediction
```

Compare:

Simple Embedding Classifier
vs
RNN Classifier

Students should explain:

Simple classifier averages word meanings.
RNN reads words in sequence and keeps hidden state.

Module 9: Simple RAG Without LLM

Build a FAQ answer system using the knowledge base.

Pipeline:

```
User Question
↓
TF-IDF Vectorization
↓
Cosine Similarity
↓
Find Best Matching KB Question
↓
Return Stored Answer
```

Example:

User Question:
Where can I watch recordings?

Matched KB Question:
Where are recordings available

Answer:
Recordings are available inside the course dashboard.

No OpenAI.

No LangChain.

No vector database.

No paid API.

Module 10: Prompt Engineering Templates

Students only create prompt templates, not execute them.

Create:

```
Zero-shot prompt
One-shot prompt
Few-shot prompt
Role-based prompt
```

Example zero-shot:

```
Classify the following support message into one category.
```

```
Categories:
```

```
Course Access, Login Issue, Certificate, Recording, Schedule, Feedback
```

```
Message:
```

```
I cannot access the course.
```

Example role prompt:

```
You are a support assistant for an online training company.
```

```
Classify the message and suggest the most suitable response.
```

Final Integrated Output

```
=====
SMART TRAINING SUPPORT INTELLIGENCE SYSTEM
=====
```

```
User Message:
```


I cannot login to the LMS

Preprocessed Tokens:

['i', 'cannot', 'login', 'to', 'the', 'lms']

Predicted Category:

Login Issue

Predicted Sentiment:

Negative

Top Similar Tickets:

1. I am unable to login to the LMS
2. I forgot my LMS password
3. I cannot access the course

Simple RAG Answer:

Use the forgot password option on the LMS login page.

Named Entities:

LMS → detected as keyword/domain term
